

Building a Private Smart Home: User Motivations and Challenges

Annika Sabrina Schulz

Open Home Foundation

Baar, Switzerland

annika.schulz@openhomefoundation.org

Paulus Schoutsen

Open Home Foundation

Baar, Switzerland

paulus.schoutsen@openhomefoundation.org

Abstract

Smart homes promise convenience, but often rely on cloud services and increased data collection. Privacy-focused people who aim to limit these dependencies may pursue privacy-preserving alternatives, often at the cost of additional effort and ongoing maintenance. We investigate privacy-oriented setup and maintenance practices among users of the open-source platform Home Assistant through 12 semi-structured interviews. Using reflexive thematic analysis, we developed three themes: (1) privacy is a driver for abandoning smart home components, (2) making the smart home private is not an end in itself, and (3) there are recurring strategies for setting up and maintaining a private smart home. We discuss implications for designing smart home platforms and tools that better support privacy-preserving configurations.

CCS Concepts

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**.

Keywords

Smart home, IoT, Privacy, Home Assistant, Open-source

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1 Introduction

Smart home technologies have become increasingly common in private households, enabling users to connect devices, automate routines, and integrate services across everyday life [6, 14]. Many commercial smart home ecosystems rely on cloud services and data collection that can track user behavior across devices and contexts. [3, 8]. People are becoming more concerned about how their personal data is collected and monetized across connected technologies, not only in smart homes [3, 8]. While privacy-preserving setups are possible, they often require substantial effort from individuals who set up and maintain a smart home. In this paper, we aim to clarify how people pursue privacy in practice and what support smart home platforms and manufacturers could provide to make privacy-preserving setup and maintenance easier.

We use the term *private smart home* to describe setups in which people aim to retain control over smart home data and reduce data sharing beyond the household. In practice, this often involves keeping device control local where possible and limiting reliance on cloud services [10, 17]. These choices are shaped by heterogeneous smart home ecosystems, where devices across brands, protocols, and apps must be combined and maintained [9, 16]. To support privacy-oriented decisions, researchers have proposed privacy labels that make privacy-relevant device properties more legible during device choice [15].

Prior research has examined smart home use in domestic contexts, including questions of usability, privacy, security, and appropriation [7, 9, 17]. For example, prior work on appropriation has shown how smart home systems are gradually assembled and evolve over time through everyday experimentation and adaptation [5]. However, privacy was not a central analytic focus in this earlier work, and smart home data practices and public awareness of data monetization have shifted considerably since then. The practical work of establishing and maintaining a private smart home, including why people value privacy, how they translate concerns into concrete setup decisions, and the challenges and trade-offs they encounter, therefore remains underexplored. This gap is particularly relevant given the tension between stated privacy concerns and the actual technology choices users make. This is commonly called the *privacy paradox* [1]. In the smart home, this mismatch can be amplified by ecosystem dependencies and interoperability demands [9, 16].

We conducted 12 semi-structured interviews with smart home users of the open-source platform Home Assistant [4] (governed by the Open Home Foundation), who described privacy as an important value in their smart home decisions. Guided by two research questions, we examine privacy-oriented setup and maintenance practices: (RQ1) What measures do smart home users take to ensure privacy in their smart home? and (RQ2) What aspects of smart home technology make users uncomfortable due to privacy concerns?

We analyzed the interview data through thematic analysis [2] and report three themes: (1) privacy is a driver for abandoning smart home components, (2) making a smart home is not an end in itself, and (3) there are recurring strategies for setting up and maintaining a private smart home. Together, these findings provide empirical insight into how privacy values translate into concrete setup decisions for end users and manufacturers of smart home technology. For smart home providers, we highlight implications for designing smart home platforms and tools that better support privacy-preserving practices over time.

2 Research Approach

We used a two-part qualitative study design consisting of a short pre-task followed by a semi-structured interview. Prior to the interview,



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participants created a list of best practices for setting up a privacy-preserving smart home based on the following scenario:

A close friend/family member/colleague is moving into their new home and finally has the chance to set up the smart home they've always wanted. They trust your advice and specifically ask how to protect privacy in their new smart home. What would you recommend that person to do? Please help us create a list of best practices for that person.

This task sensitized participants to privacy-related decisions and served as an interview prompt, similar to context-mapping approaches that combine preparatory activities with follow-up conversations to help participants recall and reflect on past experiences [13]. Participants were recruited from an existing research pool of Home Assistant users who signed up to be contacted for future studies. We conducted 12 interviews with participants from multiple countries (primarily in Europe, as well as the United States and Nigeria) in English and German. Two of the 12 were pilot interviews; since the interview guide did not change afterwards, we included them in the analysis. All participants identified as men and lived in shared households.

We conducted interviews via video call following a semi-structured guide about each participant's household context, smart home setup, and privacy-related decisions during setup and maintenance. At the beginning of each interview, we asked participants to rate their smart home privacy on a 0-10 scale (0 = not private, 10 = highly private).

We analyzed the data using reflexive thematic analysis (RTA) [2]. We chose a reflexive approach because our goal was to develop interpretative themes that capture participants' accounts while acknowledging the researcher's active role in shaping the analysis. RTA conceptualizes themes as analytic outputs generated through engagement with the data, rather than as patterns that exist independently of interpretation. The approach was primarily inductive and proceeded iteratively through familiarization, coding, theme development, and refinement.

The analysis was conducted by the first author, a German HCI researcher with expertise in the privacy paradox, deceptive design patterns, smart home market practices, and household dynamics in shared smart homes. The Open Home Foundation is a non-profit organization whose mission and core values emphasize improving privacy and user choice in smart homes. Privacy is therefore not only a research topic but also a practical and normative concern in the authors' everyday work, shaping what the researcher attends to in the data and serving as an explicit interpretative lens.

Transcripts were automatically generated and manually corrected in Dovetail, where coding was conducted. The researcher repeatedly revisited transcripts and audio/video recordings to deepen familiarity. Initial codes were grouped into candidate themes based on shared meaning and iteratively refined, collapsed, or expanded through comparison with the full dataset. The process was recursive and included ongoing reflection on how analytic decisions and the researcher's standpoint shaped the themes constructed.

3 Findings

Ultimately, we created three themes that capture how participants approached privacy-oriented smart home setup and maintenance.

3.1 Theme 1: Privacy drives abandonment and replacement of smart home components

Participants described privacy as a key driver for removing, replacing, or disengaging from smart home components, including devices, services, and entire ecosystems. These decisions were often framed as responses to perceived loss of control, cloud dependence, and concerns about vendor lock-in. For several people, privacy was not a fixed requirement but an evolving expectation: technologies that once felt acceptable were later reconsidered as individual risk perceptions and awareness of data practices changed.

P5 described how dependence on a single commercial account gradually eroded trust and motivated moving away: *"the fact that all my life is tied to one single account that I don't have control over [...] started eroding my trust in Google."* Participants also expressed that their trust in privacy promises remained conditional and could shift quickly if companies changed their business models. P7 captured this uncertainty: *"we'll see for how long [companies] can do that."*

Importantly, privacy perceptions were not only shaped by abstract concerns but could shift in response to events in a user's personal life. Such events included the birth of a child, another person moving into the household, or changes in politics that made data practices feel more consequential. For example, P2 described removing a smart speaker from the bedroom after discussing privacy when his wife moved in, noting that he *"never missed it."*

Concerns became even more pronounced when children were involved. P5 explained that after having a daughter, cloud-based photo services felt riskier. He gave the example of human review of pictures in the cloud: *[My wife and I] talked especially because we have a small daughter [...] there was this big thing with both Apple and Google going through the photos on Google Photos and Apple Photos because of like child abuse - and that's great! But the problem is, there were false positives, and those photos are being seen by humans, [...] do I really need a random human going through my pictures if they are possibly flagged for some reason?* Participants also anticipated privacy risks emerging from new technological developments that were not foreseen when devices were first installed. In particular, they expressed uncertainty about future data uses such as large-scale aggregation and the training of AI systems on user-generated data. This uncertainty contributed to a precautionary stance in which avoiding data exposure was seen as easier than managing consequences later.

Beyond household changes, participants linked privacy-oriented abandonment to broader political developments and shifting power relations between technology companies and states. P5 argued that large platforms increasingly resemble political actors: *"big tech companies are more and more powerful to a point where they kind of dwarf some smaller nations"*.

3.2 Theme 2: Making the smart home private is not an end in itself

When asked to rate the privacy of their smart home setup on a 0-10 scale, participants reported a moderate average rating ($M=7.2$, $SD=1.7$). While they expressed confidence in their setups, they also acknowledged that some devices and services remained less private. This included, for example, keeping cloud-dependent services and devices for convenience and functionality, resulting in "good

enough” privacy rather than complete control. Across interviews, participants framed privacy as a continuous negotiation rather than a final state: they aimed to reduce exposure, but did not expect to achieve a fully private, fully local, or fully secure smart home. P9 summarized this perceived limit in connected systems: *“the only way to be 100% secure is to be disconnected.”*

Participants emphasized that privacy was not among their initial motivations for adopting smart home technologies. Instead, they typically started with functional goals such as automation, convenience, or monitoring, and only later began aligning their setup with privacy expectations. As a result, privacy-oriented decisions often required concrete trade-offs that were not always obvious upfront. For example, disconnecting devices from the cloud can disable features that vendors tie to online accounts, such as advanced configuration options or specific device controls. P4 described how blocking cloud communication directly reduced functionality: *“just blocking that stuff and then stuff stops working because [...] all of a sudden you can’t control the colored parts of the light”*. This made privacy-preserving choices feel like accepting a “reduced” product experience, even when the hardware itself was technically capable. Participants described weighing these losses against the benefits of keeping data local and reducing dependence on external platforms. P7 captured this tension: *“it’s functionality versus [...] protecting privacy as a trade off [...] at some point, I’m like: Ahh, I don’t care.”*

Participants also described privacy work as effortful and often unrewarding, particularly when data flows were difficult to inspect or control. P7 described his experience: *“you don’t know what is being sent out [...] there’s encryption [...] I don’t have the time [...] the will [...] and I don’t have the knowledge.”* Similarly, P1 emphasized that deeper network reconfiguration could be a lot of work with uncertain benefit: *“I could do an awful lot [...] messing around with the network architecture [...] but I’m sort of afraid that I’m not going to gain that much value out of that activity.”*

Finally, participants described privacy choices as not only technical but also social. Several described privacy as part of everyday conversations: when showing their setup to friends, discussing devices with other smart home enthusiasts, or justifying why certain products were avoided. In these moments, participants expressed feeling judged for being “too strict” or “too paranoid”. P7 emphasized the desire to be seen as reasonable: *“I don’t want to be tinfoil hat”*. P5 noted that even among technically savvy peers familiar with his setup, privacy-oriented practices could be seen as excessive: *“they actually think I go a step too far”*. This suggests that maintaining a private smart home also involves managing how privacy values are explained and defended to others, not only how devices are configured.

3.3 Theme 3: There are recurring strategies for setting up and maintaining a private smart home

Across interviews, participants described a set of recurring strategies for building and maintaining a privacy-preserving smart home. Because such decision-making requires weighing different technological milieus, we hypothesize that such privacy-preserving strategies may be even more difficult to pursue for less technologically confident users. A central strategy was to prioritize setups

perceived as “local” and less dependent on external platforms. Participants frequently preferred open protocols (e.g., Zigbee) and avoided Wi-Fi devices when possible. At the same time, participants were cautious about oversimplified advice and questioned whether protocol choice alone was sufficient. P1 reflected on common recommendations to “only use Zigbee”: *“really? I’m not so sure that really solves the problem, Wi Fi versus Zigbee.”*

Participants also described developing heuristics for deciding which devices to trust. Several were skeptical of products that seemed unusually cheap, even when they used open protocols. P4 captured this suspicion: *“it’s crazy cheap [...] like if you guys came out and your [device] was like \$12 I’d be concerned.”*

Another strategy involved network segmentation, especially for Wi-Fi devices. Participants described trying to isolate smart home devices on a dedicated network, but also noted uncertainty about whether this effort fully addressed privacy and security concerns. P9 reasoned through this trade-off while comparing options: they *“personally prefer to use non-WiFi”* (e.g., Zigbee, Z-wave, Matter over Thread), and described Wi-Fi as *“probably the least secure”* option *“unless you do go so far as to segment it out”*. However, participants emphasized that these strategies were difficult to apply consistently in everyday life. Even when they aimed to place smart home devices on a separate IoT Wi-Fi network - kept apart from personal devices and sometimes restricted from accessing the broader home network or the internet - simple needs could lead to shortcuts. P4 described: *“If you use a closed VLAN for your devices, they don’t get auto-discovered sometimes. I just need a red light bulb [...] I hooked it into the main network, and it instantly undoes all of the work of having a really private network, and then I forget about it. Six months later, I’m like: oh, that light bulb is not on the smart home network[...] I’m not gonna fix it because I’m just tired.”*

This gap between “ideal strategies” and everyday realities was also visible in how participants approached the best-practice advice task. When giving recommendations to a fictional friend setting up a private smart home, participants described strategies that were systematic and proactive, emphasizing that privacy-preserving choices should be made early and planned in advance of choosing devices. In contrast, their smart homes often evolved gradually over time, shaped by incremental additions, changing needs, and accumulated maintenance work. This made it difficult for some participants to consistently follow privacy goals.

In this context, “starting from scratch” became a viable strategy for regaining control: participants described removing components of their smart home and rebuilding the setup from the ground up. This process allowed them to revisit earlier choices, replace or remove services and devices, and realign the system with their current privacy expectations. Participants did not necessarily rebuild their smart home *because* of privacy concerns, but often used opportunities, such as moving homes, to make privacy-oriented changes more systematically. P5 shared: *“I did my research for like two years... I had talks with my wife back and forth... and then I started everything from scratch”*. Similarly, referring back to the 0-10 privacy scale discussed earlier, P4 described: *“I would say I started at a one, open door policy [...] But as I’m getting ready to rebuild the whole system... I’m gonna use a lot of the stuff, but certainly a fair number of things are getting removed because I have the opportunity to move on from them. I’m aiming for like a strong eight or nine”*.

4 Discussion and Conclusion

Our findings show that building a privacy-preserving smart home is not only a matter of initial device selection but also an ongoing practice shaped by shifting privacy expectations, persistent trade-offs, and strategies that require sustained effort. The participants highlighted that efforts to build a smart home are frequently incremental, occasionally exhausting, and often difficult to maintain consistently, even among technically experienced users. This suggests a design opportunity for smart home platforms to better support privacy as a longitudinal, maintenance-oriented activity rather than a one-time setup decision.

Enabling privacy reassessment in changing contexts Prior work has argued that privacy in the smart home is not a fixed state but is continuously renegotiated as technologies, practices, and boundaries evolve [17]. Consistent with this, participants' privacy practices often shifted when personal circumstances changed (e.g., a partner moving in, the birth of a child, or changing routines). These reassessments were not only individual: privacy in the smart home is relational and embedded in shared domestic life, where preferences and boundaries may differ between household members and visitors [7]. This suggests that privacy-preserving smart home initiatives should support not only technical configuration, but also reflection on and communication of evolving privacy preferences.

Beyond household dynamics, participants also explained that political developments and broader news about data misuse shaped what felt acceptable or risky. Notably, participants described staying informed about privacy-relevant developments, such as breaches, policy changes, vendor acquisitions, and political events that can shift what data practices are considered acceptable. This suggests that people who are less engaged with technology news may miss important triggers for reassessing their setup, and that platforms and organizations could help by surfacing relevant updates when affected devices, integrations, or vendors are involved.

Supporting privacy as an ongoing, visible, and actionable practice Multiple participants expressed uncertainty about what data is being collected and transmitted, as well as fatigue from having to continuously monitor and adjust configurations. This connects to privacy research that has long highlighted tensions between the stated concerns of users and their actual technology choices, commonly known as the *privacy paradox* [1]. In the smart home, these tensions may be amplified by ecosystem dependencies and opaque data flows, making it difficult for users to align intentions with actual system behavior. Several participants suggested that smart home platforms could better support privacy maintenance by making privacy status more visible and actionable, for example, by initiating regular reviews of privacy-relevant settings and data flows, such as a periodic "privacy checkup" that helps users revisit earlier decisions and detect unintended changes.

Motivating and guiding privacy-oriented action Participants emphasized that privacy improvements are rarely rewarding in themselves and often compete with more urgent household needs, suggesting that users may need motivational support and low-effort prompts to follow through on privacy-related maintenance work. Participants proposed that platforms could provide an understandable assessment of "how private" a smart home currently is, paired with concrete, prioritized actions to improve it. Others suggested

more proactive guidance during device selection, such as recommendations for privacy-respecting devices or having access to a trusted label to reduce reliance on ad hoc heuristics. Interactive privacy labels have been proposed in smart home research to make privacy-relevant device properties more legible and actionable [15]. Together, these suggestions point to opportunities for smart home platforms to support not only awareness of privacy issues, but also sustained action over time.

Making data flows legible through network-level feedback

The strategies participants used often relied on limiting outbound connections, segmenting networks, or avoiding cloud dependencies, but these efforts were difficult to verify and maintain. A recurring suggestion was to visualize the degree to which a smart home system "talks to the outside," for example, by showing which devices or integrations communicate externally, when, and for what purpose. Participants also expressed interest in broader network-monitoring support that could flag privacy or security vulnerabilities across smart home networks. Such feedback could help users detect unexpected connections early and reduce the burden of manual investigation. In addition, the uncertainty among participants around protocols and device behavior suggests that platforms could provide educational content on protocol-level privacy implications and system inner workings, helping users feel more in control and better able to make informed privacy decisions.

4.1 Limitations and Future Work

The participants for this study were recruited from the Home Assistant community, an open-source platform often positioned as a privacy-oriented alternative to commercial smart home ecosystems. However, the research questions focused on how people reason about and enact privacy-preserving smart home practices more broadly, and the findings do not refer to Home Assistant-specific features. All participants identified as men. While this reflects the gender distribution within the Home Assistant community and the gendered divisions of smart home user roles [11, 12], it limits the perspectives represented in our dataset. All participants set up and maintained their household's smart homes and described themselves as highly technical and privacy-aware while doing so, offering insight into expert practices but potentially limiting transferability to less technical households. Participants were primarily based in Europe and the United States, limiting cultural and regulatory diversity. Future work should explore how privacy-preserving smart home practices and tools can be made more accessible and maintainable across a broader range of households and platforms.

5 Conclusion

We investigated how privacy-oriented smart home users set up and maintain their smart homes. Our findings show that privacy concerns can drive people to abandon components, and that privacy involves ongoing trade-offs rather than a final state. Participants described strategies for moving toward more private setups, yet also paradoxical behavior where privacy goals compete with convenience and effort, making privacy difficult to sustain. These insights highlight opportunities for smart home platforms to support privacy as an ongoing, actionable practice through transparency, guidance, and reduced maintenance burden. More broadly, they

suggest design opportunities to better support privacy and user control across ubiquitous connected technologies.

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